

Manasa Vijayaraghavan

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github.com/Manasavijr | Portfolio

EDUCATION

University of Southern California	Los Angeles, CA
M.S. Applied Data Science	May 2027
SRM Institute of Science and Technology	Chennai, India
B.Tech. Computer Science Engineering	Jun 2025

EXPERIENCE

AI Native Intern — Wipro	June 2026 – Aug 2026
- Built agentic decision-support workflows for life insurance using Python, LangGraph, and LLMs, orchestrating multi-step data gathering, validation, reasoning, and conditional routing across underwriting and fraud/contestability use cases. - Owned Life Underwriting Intelligence and Life Fraud & Contestability Review workflows end to end, modeling processes as stateful LangGraph nodes with human-in-the-loop approval gates before consequential decisions. - Integrated agent workflows with enterprise systems including Guidewire, ServiceNow, and Majesco, enabling structured retrieval and exchange of operational data across insurance workflows.	
Data Science Intern — Ford Motor Company	Feb 2025 – May 2025
- Built data pipelines in Python: ingested, cleaned, and structured raw multimodal operational data across multiple formats; ran exploratory analysis to surface patterns and anomalies before downstream processing; documented data provenance and processing steps; deployed on GCP, improving efficiency by 75%. - Wrote clean, documented, reproducible Python code; collaborated with engineering teams; demonstrated ability to work independently with increasing autonomy.	
Data Science Intern — iNextLabs	Aug 2024 – Oct 2024
Built NLP data pipelines processing unstructured text data using Python and Pandas/NumPy; applied anomaly detection on operational datasets; documented processing steps for reproducibility; delivered production data solutions to 500+ users (40% improvement).	

PROJECTS

PII Detection & Privacy-Preserving Data Pipeline	(Python, spaCy, NLTK, Pandas, NumPy, Regex, Differential Privacy (ϵ -DP), Flower (flwr), PyTorch, Git)
- Built PII detection and removal pipeline using spaCy NER and custom Regex pattern matching: detected and anonymised names, emails, phone numbers, and financial identifiers across unstructured text datasets; implemented anonymisation workflows (masking, pseudonymisation, redaction); applied differential privacy (ϵ-DP, Gaussian noise) for privacy-preserving data processing across 10+ distributed nodes. - Documented data provenance and processing steps for every dataset through the pipeline; wrote clean, reproducible Python scripts; tested anonymisation workflows end-to-end.	
Data Quality Scoring & Multi-Format Pipeline	(Python, PySpark, SQL, Pandas, NumPy, SciPy, XGBoost, Isolation Forest, MLflow, GCP, Git)
- Ingested, cleaned, and structured raw operational datasets across multiple formats on 1M+ rows: multi-format data ingestion, schema validation, and quality checks; developed data quality scoring models (XGBoost 87% AUC) assessing dataset value; ran exploratory analysis surfacing patterns, anomalies (Isolation Forest), and coverage gaps before downstream processing. - Built automation scripts improving pipeline efficiency; documented data processing steps and assumptions with MLflow experiment tracking.	
SignMate AI	(Patent Filed — Python, PyTorch, TensorFlow, OpenCV)
Patent filed for a production AI system — clean, documented, reproducible code shipped end-to-end with real-world impact.	

TECHNICAL SKILLS

Privacy	PII Detection, Anonymisation, Differential Privacy (ϵ -DP), spaCy NER, Regex
Agentic AI	LangGraph, Human-in-the-Loop, Anthropic Claude API, LangChain
Enterprise Systems	Guidewire, ServiceNow, Majesco
Pipelines	Python, Pandas, NumPy, PySpark, SQL, Data Quality Scoring, Multi-Format Ingestion
NLP	spaCy, NLTK, HuggingFace, Regex, Text Cleaning, NER, Deduplication
Analysis	EDA, Anomaly Detection, Isolation Forest, MLflow, GCP, Git
Recognition	Patent Filed (SignMate AI)